

Arnesh Dorsatwar

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Engineering

Atlanta, GA

Expected May 2028

EXPERIENCE

FigTree IT (Fireweave)

Software Engineering Intern

Remote

Oct 2025 – Present

- Developed features for **Fireweave**, a network security policy management platform for automating firewall policy analysis and traffic flow optimization.
- Built **Ember**, a **distributed LLM-based interface** to manage FireWeave tools, supporting chat persistence, multi-step workflows, and context-aware reasoning.
- Fine-tuned ollama 3.1:8b using **Unsloth + QLoRA**, implementing knowledge distillation and dataset normalization pipelines, achieving **53% faster inference** compared to the original models for Network Security.
- Implemented **Adaptive RAG pipelines with LangGraph** for Ember to query live firewall configs, policy changes, and enterprise rules and include support for external documents in real time increasing speed by 22%.
- Migrated backend storage from a document database to **PostgreSQL**, introducing Redis caching to reduce query latency and improve throughput.

BestBrains

Computer Science Teacher

Phoenix, AZ

Apr 2025 – Oct 2025

- Instructed **50+ students in Scratch, Python, and HTML**, focusing on project-based learning.
- Designed and launched a **summer program** that attracted nearly 30 new students.
- Developed a new **computer science curriculum** with game design principles for **100+ students**.

ArneshKeys

Founder & Lead Designer

Phoenix, AZ

Apr 2022 – Present

- Founded and managed a custom mechanical keyboard business serving international clients.
- Built **over 40 keyboards** and designed **20+ unique PCBs** with advanced circuit and CNC integration.

Willow TV

Broadcasting Assistant

Phoenix, AZ

Jul 2024 – Dec 2024

- Edited cricket footage and created broadcast montages under the production director's guidance.

PROJECTS

Homelab Media Server — Self-Hosted Infrastructure | Ubuntu Server, Docker, Networking

- Built a **24/7 Ubuntu server** from salvaged hardware, performing diagnostics (capacitor repair, CPU swap) to restore a non-POST system.
- Deployed a **Docker Compose stack** (Jellyfin, Sonarr, Radarr, Home Assistant) with health checks and auto-restart.
- Configured **Tailscale VPN** for secure remote access without public IP exposure.
- Managed **storage, transcoding, and GPU passthrough** on a headless SSH-based system.
- Monitored **system performance** (CPU, RAM, I/O, network) for stable multi-user uptime.

Cow GPS Tracker — Embedded IoT Monitoring System | ESP32-C6, Embedded Systems, Flask

- Built a **real-time telemetry system** using ESP32-C6 over 915 MHz FSK, tracking RSSI and packet delivery.
- Developed a **Flask dashboard** with live mapping, geofencing, and anomaly detection.
- Debugged **SPI, UART, I2C** issues, resolving signal loss, firmware mismatches, and power faults.
- Implemented **adaptive reporting (1–60s)**, improving battery life and reducing traffic up to 60x.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#

Frameworks & Tools: Node.js, Heroku, Railway, AWS, MongoDB, PostgreSQL, Redis, LangChain, RAG, Unsloth, QLoRA, Cursor, AutoCAD, Palo Alto, SFT, Pytorch

Developer Skills: Git, Linux, REST APIs, CAD Modeling, PCB Design Docker, Media Automation (Arr Stack)